

TORPSHAMMAR, Sweden

WMO: 023430

Lat: 62.494N

Lon: 16.277E

Elev: 328

StdP: 14.52

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.3	(c) -8.6	(d) -19.2	(e) 1.9	(f) -14.1	(g) -13.0	(h) 2.7	(i) -8.5	(j) 12.3	(k) 35.6	(l) 10.5	(m) 33.9	(n) 0.3	(o) 280	(p) 0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	80.4	64.2	76.6	62.4	73.0	60.7	66.7	75.7	64.6	73.3	62.5	70.2	4.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	88.3	70.0	61.2	81.8	67.8	59.4	76.5	65.9	31.4	75.7	29.8	73.3	28.3	70.3	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.2	8.7	-19.9	84.5	5.3	3.3	-23.8	86.9	-26.9	88.9	-29.9	90.7	-33.7	93.2
			-19.6	69.8	4.9	2.5	-23.2	71.6	-26.0	73.1	-28.8	74.5	-32.4	76.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.7	17.3	19.7	28.4	38.1	47.8	55.3	60.6	57.9	49.2	38.3	29.0	21.0		
	DBStd	16.99	12.27	11.66	8.79	5.59	6.51	5.52	5.28	5.50	5.94	6.56	8.27	12.15		
	HDD50	5006	1013	848	669	357	122	13	1	5	85	364	631	898		
	HDD65	9645	1478	1268	1134	806	535	294	159	228	473	826	1081	1363		
	CDD50	869	0	0	0	1	52	172	329	250	62	3	0	0		
	CDD65	33	0	0	0	0	0	3	22	8	0	0	0	0		
	CDH74	612	0	0	0	0	36	133	332	111	0	0	0	0		
	CDH80	112	0	0	0	0	2	19	79	12	0	0	0	0		
(14) Wind	WSAvg	2.9	2.1	2.5	3.4	3.5	3.9	4.2	3.3	2.6	2.7	2.7	2.3	2.1		
(15) Precipitation	PrecAvg	25.4	1.7	1.1	1.2	1.5	1.7	2.6	3.6	3.5	2.2	2.4	2.0	1.9		
	PrecMax	35.9	3.4	2.2	2.6	4.1	4.3	4.5	9.0	6.6	6.6	6.2	5.1	3.1		
	PrecMin	19.3	0.2	0.1	0.2	0.1	0.4	1.1	0.6	1.8	0.6	0.7	0.7	0.2		
	PrecStd	4.0	0.9	0.5	0.6	1.0	0.9	1.0	1.9	1.5	1.4	1.3	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	45.6	52.8	67.2	78.5	82.7	86.7	81.9	70.9	60.1	50.4	44.4		
		MCWB	39.1	39.3	43.8	52.0	59.4	62.7	67.0	65.9	59.9	53.4	47.7	40.7		
	2%	DB	38.5	42.4	47.8	59.5	72.8	78.1	82.0	77.2	66.8	54.9	45.8	40.9		
		MCWB	35.6	37.9	40.3	47.5	56.4	61.0	65.6	64.7	58.0	50.1	43.7	38.1		
	5%	DB	35.3	39.0	44.5	55.3	67.6	73.3	77.6	73.2	63.8	51.9	42.2	37.9		
		MCWB	33.4	35.5	38.2	44.8	54.0	58.8	64.0	62.7	56.5	48.6	40.6	36.0		
	10%	DB	32.7	34.9	41.2	51.2	62.8	69.1	73.7	69.3	60.3	49.3	39.7	35.0		
		MCWB	31.6	32.6	36.1	42.7	51.9	57.0	62.1	60.9	54.6	46.3	38.2	33.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	40.0	44.5	53.3	61.1	65.9	70.0	68.8	61.7	54.7	48.3	41.2		
		MCDB	42.2	44.1	52.3	67.0	74.4	77.9	80.5	76.5	68.9	58.6	50.2	43.2		
	2%	WB	36.1	38.0	41.1	48.5	58.2	62.1	67.4	66.3	59.3	51.2	43.9	38.5		
		MCDB	37.9	42.0	47.1	57.8	70.1	72.5	77.3	73.6	65.1	53.5	45.8	40.6		
	5%	WB	33.4	35.8	38.7	45.9	55.2	60.1	65.5	64.1	57.3	49.1	41.0	36.0		
		MCDB	34.7	38.8	43.7	53.9	65.8	70.2	75.0	71.0	62.0	51.6	41.9	37.6		
	10%	WB	31.7	32.8	36.4	43.2	52.6	58.1	63.4	62.1	55.3	46.5	38.3	33.7		
		MCDB	32.8	34.7	40.7	50.4	61.6	67.2	72.0	68.1	59.7	49.0	39.4	34.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	15.9	18.8	20.5	22.3	21.5	21.1	19.3	18.6	14.1	10.2	12.3		
		MCDBR	13.9	16.5	21.3	29.8	32.5	30.5	28.6	25.6	23.1	19.0	13.1	14.8		
		MCWBR	12.3	12.7	14.7	18.2	17.8	15.7	14.3	14.0	14.9	14.5	11.7	12.8		
	5% WB	MCDBR	12.8	16.1	20.2	27.1	29.1	26.7	25.2	22.4	21.1	17.8	12.6	14.4		
		MCWBR	11.8	12.6	14.3	17.3	16.9	14.5	13.4	13.1	14.2	14.2	11.6	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.288	0.314	0.351	0.355	0.349	0.365	0.357	0.323	0.308	0.248	0.347		
	taud		2.092	2.233	2.313	2.371	2.425	2.469	2.442	2.460	2.542	2.454	2.178	2.648		
	Ebn at Noon		139	211	252	265	274	278	269	261	251	207	132	87		
	Edh at Noon		10	20	26	31	32	31	31	28	22	16	9	6		
(44) All-Sky Solar Radiation	RadAvg		64	266	732	1200	1510	1743	1589	1208	759	333	95	34		
	RadStd		8	42	79	101	155	115	176	102	93	58	18	3		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page